

Day-ahead wind speed forecasting using f -ARIMA models

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ARTICLE INFO

Article history:

Received 27 March 2008

Accepted 3 September 2008

Available online 25 October 2008

Keywords:

Wind speed forecasting

Time series analysis

f -ARIMA models

ABSTRACT

With the integration of wind energy into electricity grids, it is becoming increasingly important to obtain accurate wind speed/power forecasts. Accurate wind speed forecasts are necessary to schedule dispatchable generation and tariffs in the day-ahead electricity market. This paper examines the use of fractional-ARIMA or f -ARIMA models to model, and forecast wind speeds on the day-ahead (24 h) and two-day-ahead (48 h) horizons. The models are applied to wind speed records obtained from four potential wind generation sites in North Dakota. The forecasted wind speeds are used in conjunction with the power curve of an operational (NEG MICON, 750 kW) turbine to obtain corresponding forecasts of wind power production. The forecast errors in wind speed/power are analyzed and compared with the persistence model. Results indicate that significant improvements in forecasting accuracy are obtained with the proposed models compared to the persistence method.

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1. Introduction

In grid interfaced wind energy systems, the variable nature of wind poses several operational challenges. Wind being a weather driven renewable resource with strong dependencies on the climate system, its variability occurs in time scales ranging from minutes through hours, days and several years. Wind speed variations in each of the time scales have corresponding implications on the operation of the utility system as indicated below.

Time scale	Implication
Minutes/hours	Regulation/load-following
Day(s)	Day-ahead scheduling/unit commitment
Months	Resource allocation
Year(s)	Economic viability

Of these, the first two time scales which correspond to short-term variations, have the largest bearing on utility scale operation while the last two which correspond to long-term variations, have planning and developmental implications. Under the short-term scales, accurate wind energy forecasts are critical to minimize (i) scheduling errors which impact grid reliability and (ii) market based (hour-ahead/day-ahead) ancillary service costs. Wind energy forecasts in turn, depend on wind speed forecasts and operating characteristics of the wind turbine. Therefore, the subject of wind

speed forecasting is becoming increasingly important and pertinent to the operation of electric utilities which integrate wind energy. For example, Basin Electric Power Co-operative, a wholesale power supplier to several cooperatives in the midwest U.S., claims that the accuracy of next-hour forecasting is currently poor (for about one thirds of the time, errors in the next-hour generation forecasts are greater than 50%) and significant improvements need to be made in this regard to reliably accommodate large shares of wind energy [1].

Broadly speaking, there are two approaches for wind speed forecasting, namely (I) weather based and (II) time series based. While the former uses hydrodynamic atmospheric models which incorporate physical phenomena such as frictional, thermal and convective effects, the latter (which is the subject of this paper) uses only historical wind speed data recorded at the site to build statistical models from which forecasts are derived.

Time series based forecasting methods rely on the classical Box-Jenkins methodology [2], which employs a general class of models such as the Auto-Regressive Moving Average (ARMA(p, q)), or Auto-Regressive Integrated Moving Average (ARIMA(p, d, q)) models to obtain forecasts. However, the simplest method of forecast is by the persistence method which does not require any model. For a given time series $\{y_n\}$, the persistence forecast is obtained by setting $y(n+1) = y(n)$, which implies that the average wind speed forecast for the next hour is simply equal to the average wind speed over the current hour. Customarily, the persistence method is used to benchmark the accuracy of a newly proposed forecasting method. Both ARMA [3,4,5] and ARIMA models [6] have been applied in the past to predict hourly average wind speeds. As alternatives, the use of artificial neural networks [7,8,9] including a method that factors

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spatial correlations of wind speeds [10] have been proposed towards obtaining improved predictions compared to the persistence forecast.

However, recent studies [11] suggest that temporal characteristics of hourly average wind speed fluctuations fall under a category of stochastic processes known as $1/f$ noise with the possibility of range correlations (LRC). Such processes are characterized by a slow decay of the autocorrelation function, or equivalently, a power law $S(f) \sim 1/f^\beta$ behavior of the corresponding power-spectrum. A detailed analysis of temporal wind speed records obtained from potential wind generation sites in North Dakota (ND) indicates that the fluctuations, in addition to possessing long range correlations [11], can also contain a spectrum of scaling exponents [12].

The ARMA and ARIMA models are traditionally very well suited to capture short range correlations, and hence have been used extensively in a variety of forecasting applications. However, an attempt to accommodate LRD using the ARMA/ARIMA models would require the inclusion of a large number of $AR(p)$, $MA(q)$ and differencing (d) parameters which would result in an expensive model. For wind farm applications, it may be advantageous to use a parsimonious model and yet retain LRD. Fortunately, it is possible to achieve this when the ARIMA model is modified by allowing d to assume fractionally continuous values in the interval $(-0.5, 0.5)$. The resulting class of models known as *fractional-ARIMA* or *f-ARIMA* models can then parsimoniously represent LRD through the single parameter d . Therefore, the central goal of this paper is to explore the suitability of *f-ARIMA* models towards obtaining more accurate next hour wind speed forecasts and as will be demonstrated later, the proposed approach leads to an average improvement of 42% [36–54%] in the forecast mean square error as compared to the persistence method. The rest of the paper is organized as follows. In Section 2, the modeling and parameter estimation pertaining to *f-ARIMA* models are described. In Section 3, the forecasting results with the proposed model are discussed. Finally, the conclusions are summarized in Section 4.

2. *f-ARIMA* model and parameter estimation

Box and Jenkins [2] introduced the $ARIMA(p, d, q)$ class of processes which, ever since, have been applied to a wide variety of time series forecasting applications. The general methodology of the Box–Jenkins approach involves (i) model identification, (ii) parameter estimation and (iii) diagnostic checking followed by forecasting. A recent paper [13] provides a nice illustration of these steps where ARIMA models are employed to predict next day electricity prices. In what follows, the details involved in each of these steps as it applies to the task of wind speed forecasting are described.

2.1. *f-ARIMA* model formulation

As remarked earlier, the fractional-ARIMA or *f-ARIMA* model arises as a special case of ARIMA processes when the differencing parameter ' d ' assumes fractionally continuous values in the range $(-0.5, 0.5)$. One of the features that distinguishes an *f-ARIMA* process from an ARIMA process is that the former is characterized by a slow decay in its autocorrelation function compared to the latter. This feature makes *f-ARIMA* models an attractive choice for data sets that exhibit long range correlations such as the wind speed records considered in the present study.

Let $\{y_t\}$ represent the time series of hourly average wind speeds. Then, an *f-ARIMA*(p, d, q) formulation for the series can be described by,

$$\phi(B)(1-B)^d(y_t - \mu) = \theta(B)\epsilon_t \quad (1)$$

where, $\epsilon_t \approx iid(0, \sigma^2)$, $d \in (-0.5, 0.5)$, and B is the backshift operator defined by $B(y_t) = y_{t-1}$. The functions ϕ, θ are polynomial functions of the backshift operator B given by

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p \quad (2)$$

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q \quad (3)$$

Since d assumes fractional values, the operator $(1-B)^d$ is defined through a power series expansion [21] as,

$$(1-B)^d = \sum_{j=0}^{\infty} \pi_j B^j \quad (4)$$

where $\pi_0 = 1$ and for $j > 0$,

$$\pi_j = \prod_{k=1}^{k=j} (k-1-d)/k, \quad j = 1, 2, \dots \quad (5)$$

Thus, the *f-ARIMA* model is completely described by p parameters $\phi_1 \dots \phi_p$ in Eq. (2), q parameters $\theta_1 \dots \theta_q$ in Eq. (3) and the fractional parameter $d \in (-0.5, 0.5)$.

2.2. Parameter estimation

After the structure of the model *f-ARIMA*(p, d, q) is defined, the estimation of parameters in the model can be performed by various techniques including the exact [14] and approximate [15] maximum likelihood estimators. A good overview of these techniques can be found in [16] and [17]. In the present study, Ox-ARFIMA (Ox) [18,19] is used to model and forecast the given time series. While Ox supports different techniques for parameter estimation, the exact maximum likelihood (EML) method is used in light of studies [14,20], which show the superiority of the EML approach over approximate likelihood methods when small/medium sample sizes are considered. For an *a priori* specified model order (p, d, q), $-0.5 < d < 0.5$, the program estimates the optimal parameters by the EML algorithm. However, since a given time series can be described by several models up to a given order, the optimal model *f-ARIMA*(p, d, q) needs to be chosen according to some criterion. Here, we create a family of *f-ARIMA* models by allowing p and q to vary in the range [0–5]. The upper limit of five for the model order was chosen keeping parsimony in mind. Out of the 36 possible models that result from this variation, the final model was chosen to be the one that resulted in the least AIC. The AIC (Akaike's Information Criterion) [22] is a metric that can be used to select an optimal model from a class of competing models. The AIC compares the accuracy of the model against its complexity measured by the number of parameters. For a model, the AIC is given by

$$AIC = -2 \ln L + 2P \quad (6)$$

where p is the number of parameters in the model and $\ln(L)$ is the natural logarithm of the maximum likelihood.

2.3. Model validation – diagnostic check

The purpose of model validation is to assess how well the model captures the given time series as described in Eq. (1). This can be done by assessing how close the error term ϵ_t (also known as the residuals) in Eq. (1) is to satisfying the assumption of an $iid(0, \sigma^2)$ process. To determine if the autocorrelations are zero, or

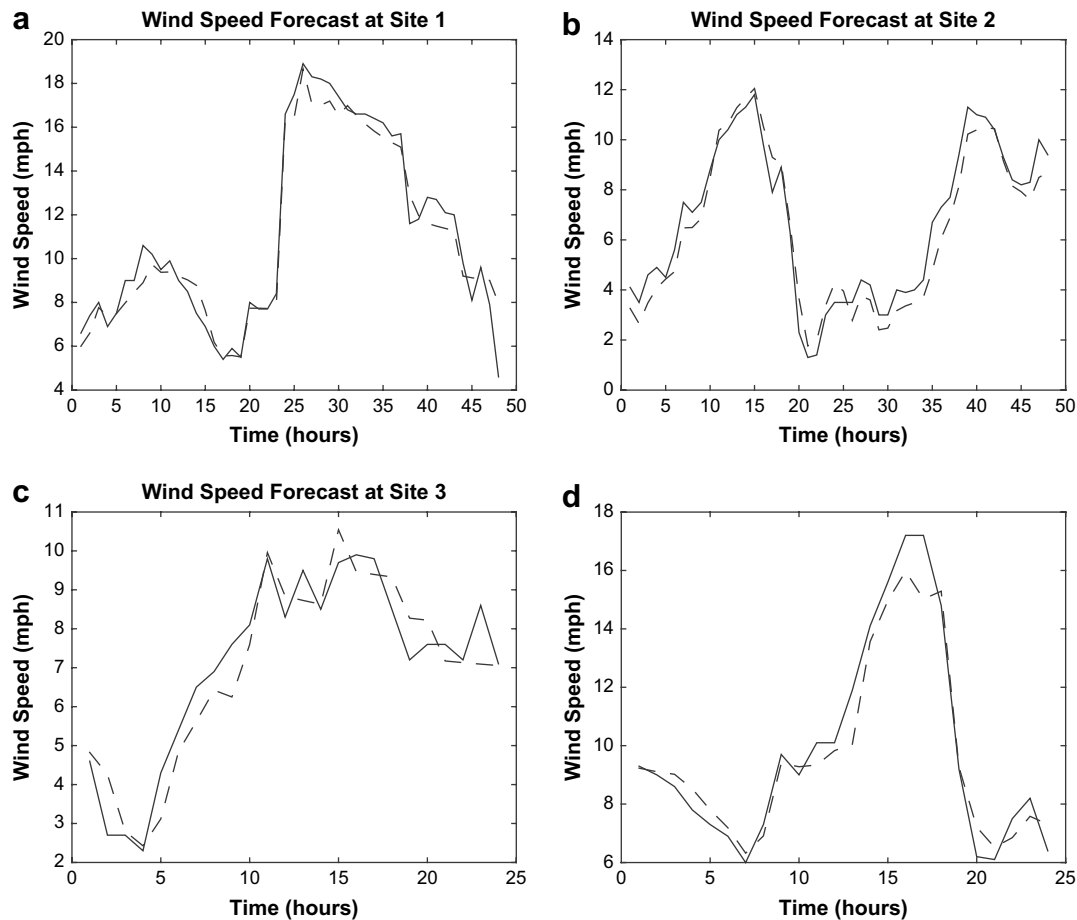


Fig. 1. Comparison of actual wind speed (solid line) with forecasted wind speed (dashed line) for sites 1–4 in (a)–(d), respectively.

insignificant after a certain lag, a simple measure of the scale of the correlations is obtained by the limits of $1.96/\sqrt{N}$ where N is the sample size. With the proposed model, we found that correlations beyond lag one are insignificant, suggesting that the residuals are sufficiently uncorrelated. Thus the models obtained are valid for the task of forecasting and these results are presented next.

3. Results

The results are organized in two subsections. First, in Section 3.1, wind speed forecasting results with the f -ARIMA models are described and the results are compared with the ARIMA and persistence based forecasts. Since the persistence model is often used to benchmark other forecasting models, the forecast accuracies obtained with the f -ARIMA models are also compared with those obtained through a neural network based model that relies on spatial correlation information [10].

Second, in Section 3.2, the wind speed forecasting results are translated into wind power forecasts using the power curve of an NEG Micon 750 kW wind turbine currently owned and operated in the region by Moorhead Public Service, Moorhead, MN. This is followed by a brief discussion on how wind speed forecasting impacts wind power forecasting, which is particularly important in the context of the day-ahead electricity market.

3.1. Wind speed forecasting results

The proposed f -ARIMA models were applied to forecast hourly average wind speeds from four wind monitoring sites in North

Dakota. A brief description of the data acquisition and site details are provided in Appendix A. In each case, four weeks of data was used to build the model. The time frames for each of the sites were selected distinctly as follows: (a) Site 1 – May, (b) Site 2 – December, (c) Site 3 – March and (d) Site 4 – October. The forecasting results for the four sites (1, 2, 3 and 4) are shown in Fig. 1(a–d), respectively.

From Fig. 1, it can be observed that the models are able to reasonably track the hourly wind speed variations at least up to a day ahead (24 h). While the models were able to maintain their accuracy over a 48 h-ahead period for sites 1 and 2, it was observed that for sites 3 and 4, the model was able to retain accuracy up to a 36 h ahead period. Therefore, the forecast is shown for a day-ahead (24 h) period for sites 3 and 4. The possible reasons for this aberration are discussed later on. The index of correlation between forecasted values and actual values for the four sites were found to be 97.38%, 98.20%, 94.46% and 98.01%, respectively, implying that on an average, the forecast produced by the model is 95.3% correlated with the actual wind speed. The forecast accuracy is further assessed by computing three indices namely (a) the Daily Mean Error (DME), (b) the variance (σ^2) and (c) the square root of the

Table 1
Comparison of forecasting methods for Site 1

Index	Persistence	ARIMA	f -ARIMA
DME (%)	45.2	144.92	33.18
σ^2	0.665	0.854	0.156
$\sqrt{\text{FSME}}$ (%)	8.43	11.87	5.35

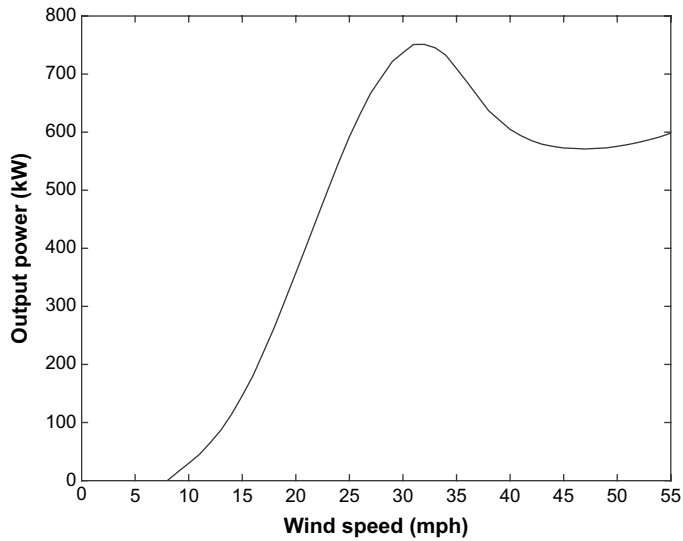


Fig. 2. Power curve of MPS's NEG MICON 750 kW WTG.

Forecast Mean Square Error ($\sqrt{\text{FMSE}}$) ($\sqrt{\sum_{i=1}^k (y_i - \hat{y}_i)^2}$) obtained with the proposed models.

The indices referred above are used to compare the forecasting abilities of the proposed method with the ARIMA and persistence based methods. The performance of these three methods are compared using these indices and a typical result (for site 1) is tabulated in Table 1. The indices obtained for the other sites are comparable.

Our results that on an average (over the four records considered) the DME is ~79.3% with the persistence model, ~117% with the ARIMA models and ~47% for the f -ARIMA models. The f -ARIMA models also yield a much smaller variance (0.24) compared to the persistence (1.07) and ARIMA method (0.89). Finally, the efficacy of the forecasts as measured by the $\sqrt{\text{FSME}}$ suggests that the proposed method yields superior forecasts compared to the other

two methods. Overall, the f -ARIMA model yields an average improvement of 42% with respect to the $\sqrt{\text{FSME}}$. While the present study considers the day-ahead forecasting horizon (24–48 h), it can be noted that a previous article [10] considers an artificial neural network (ANN) based model that relies on spatial correlations to obtain forecasting accuracy improvements of about 25% (compared to persistence method) on time scales up to 2 h. Next, the wind speed forecasting results are utilized to obtain wind power forecasts based on the power curve of an operational wind turbine.

3.2. Wind power forecasts

Moorhead Public Service (MPS), a consumer owned utility is located in Moorhead on the Minnesota–North Dakota border. MPS currently owns and operates two 750 kW NEG MICON wind turbine generators (WTG) at a hub height of 56 m. The turbines have a cut-in speed of 9 mph, a cut-out speed of 56 mph while peaking peak power at about 32 mph. The power curve of the WTG is shown in Fig. 2.

The wind speed data sets (recorded at a height of 20 m) were adjusted through the power law ($v/v_r = (z/z_r)^\alpha$ [23,24]) with the exponent $\alpha = 1/7$ to reflect wind speeds at the hub height of 56 m. The scaling relation described above is also referred as the one-seventh power law and often used to carry out vertical wind speed extrapolations in wind engineering and atmospheric pollution studies. Using the power curve data, the wind speed data sets were mapped to obtain corresponding wind power data sets by cubic interpolation. The results obtained for all of the sites (1–4) is shown in Fig. 3. The error in estimated WTG output power with the proposed model is shown in Fig. 4.

3.3. Discussion – wind speed prediction

Two important issues in any prediction scheme are (i) how well a model retains its accuracy over a forecast horizon and (ii) how robust is the scheme to the choice of the forecast horizon. To address the first issue, an initial forecast horizon of 24 h, suitable

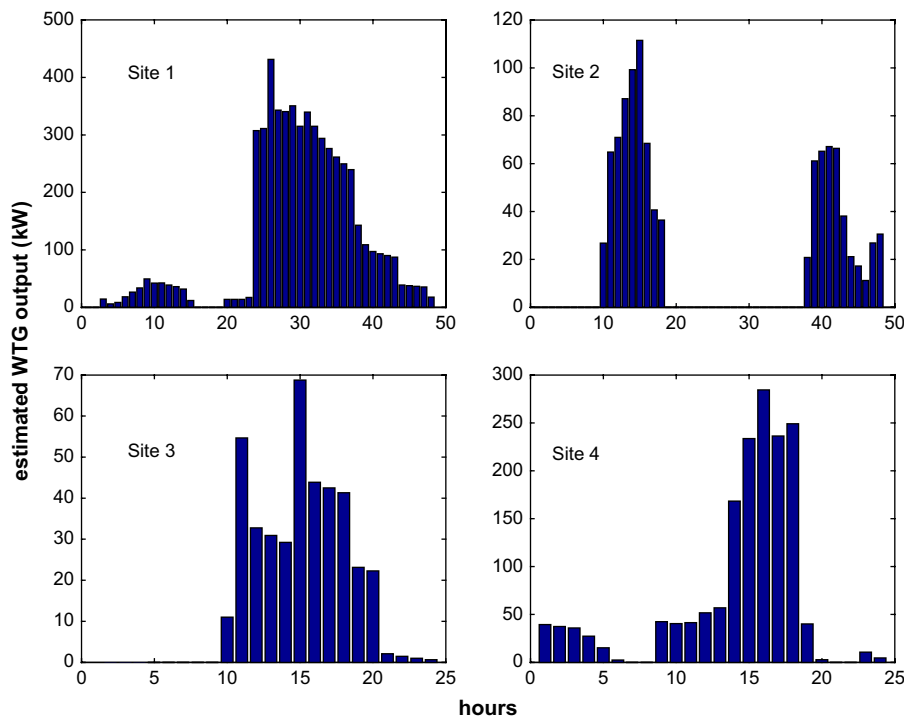


Fig. 3. Estimated WTG power output (kW) for sites 1–4.

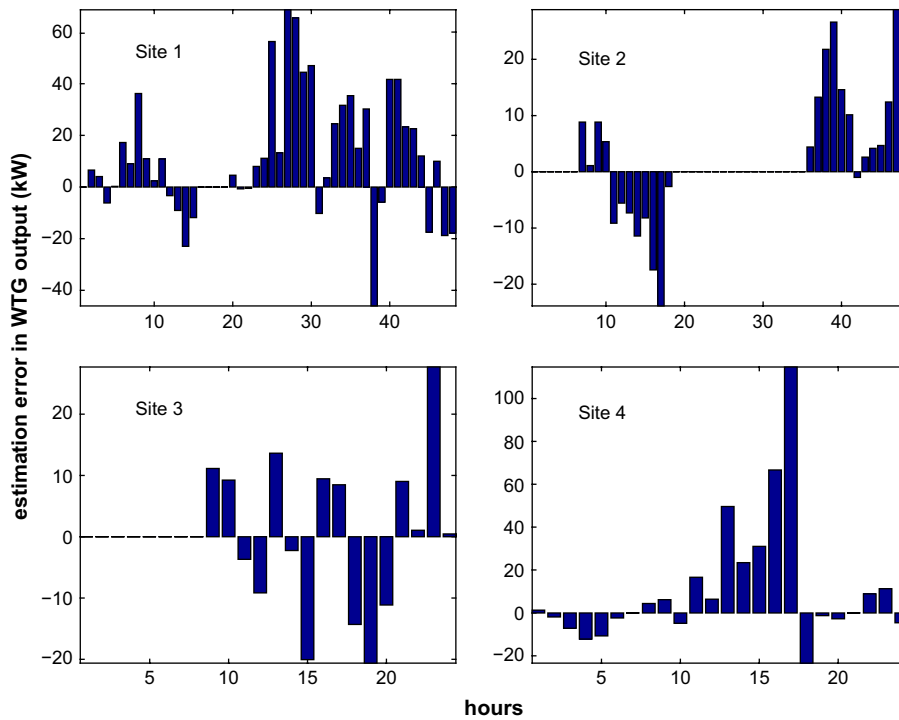


Fig. 4. Estimation error in WTG output power obtained with f -ARIMA model.

for the day-ahead market was chosen. The second issue is particularly important in the present context because wind patterns dependent on atmospheric conditions can change with seasons. To assess how robust the methodology is to the choice of the time frame, four distinct weeks, each corresponding to a site were chosen. For sites 1 and 2, the time frames correspond to June (summer) and December (winter), respectively. For sites 3 and 4, the time frames were October (fall) and March (early spring), respectively.

As the results indicate, the models yield reasonably accurate forecasts over a 48 h period for site 1 and site 2, Fig. 1(a), (b). However, the range of accurate forecast is reduced to 24 h for site 3 and 4, Fig. 1(c), (d). A possible reason for this observation is that stabler wind regimes result during summer and winter months while gustier and volatile wind patterns prevail during late autumn and early spring. Therefore, in the former case (sites 1 and 2), a single model is seen to produce the forecast over the 48 h period as compared to the 24 h period for the latter (sites 3 and 4).

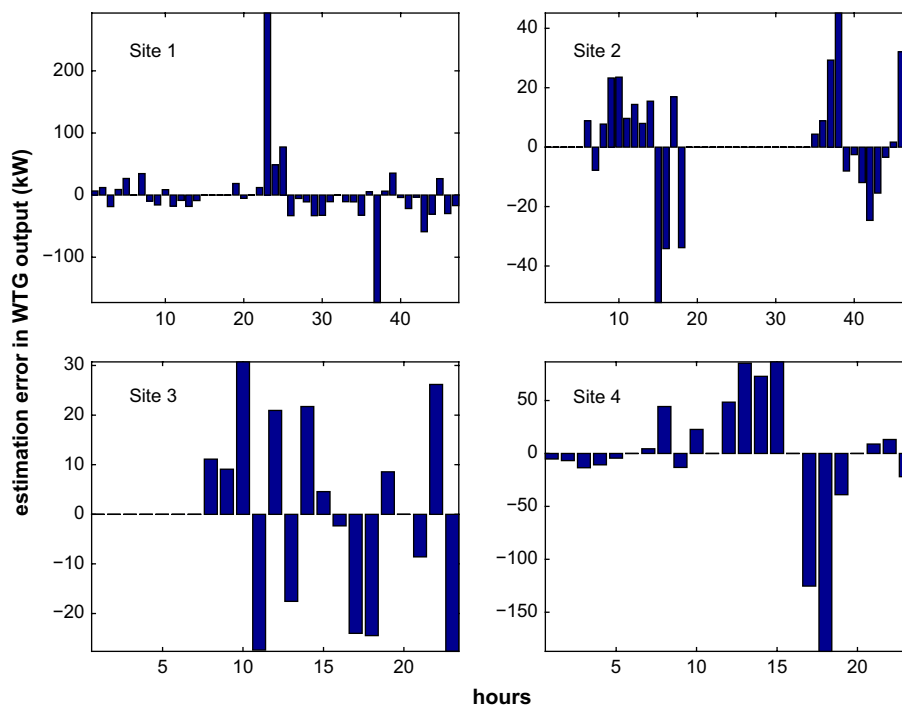


Fig. 5. Estimation error in WTG output power obtained with the persistence model.

Nevertheless, the models are seen to substantially improve the forecasting accuracy compared to the persistence forecast for a forecasting horizon of 24 h.

3.4. Discussion – wind power prediction

First, it can be observed from Fig. 3 that the estimated WTG output powers in general, mimic the temporal wind speed patterns of the corresponding sites, except when the wind speeds fall below the cut-in speed of 9 mph. In this study, none of the adjusted wind speed data sets exceed 22 mph. Since the power curve in Fig. 2 is approximately linear for wind speeds in the range of 9–22 mph, it follows that the corresponding wind power variations scale linearly with wind speeds in this region. When the wind speeds fall below the cut-in speed, for example, during [18–37] h for Site 2 and [0–10] h for Site 3, the WTG output powers are zero. Therefore, forecast errors in wind speed regimes below the cut-in speed are annulled.

Next, the efficacy of the wind speed prediction models to obtain estimates of the corresponding wind power production is studied. The estimation error in this case is defined as the difference between the WTG output obtained with the actual wind speed and that obtained with the prediction models (*f*-ARIMA, persistence). Therefore, positive error values would imply that the model underestimates, and negative values indicate that the model overestimates the actual power production. While Fig. 4 depicts the errors obtained with the *f*-ARIMA model, Fig. 5 depicts those obtained with the persistence model, for each of the sites. A comparison of Fig. 4 and Fig. 5 for the corresponding sites reveals an interesting feature. Consider the time interval [0–10] h for Site 1 when wind speed excursions [6–10 mph] (refer Fig. 1) are small. In this case, the persistence model is seen to yield smaller errors compared to the *f*-ARIMA models. However, when the wind speed exhibits a marked deviation from ~8 to ~18 mph at 22 h, the persistence model underestimates the power production by nearly 293 kW, compared to 70 kW by the *f*-ARIMA model. Conversely, when the wind speed exhibits a sharp decline at 37 h, the persistence model overestimates power production by 170 kW compared to 45 kW by the *f*-ARIMA model. A similar feature can be noticed for Site 4 at 17 h, when the wind speed exhibits a sharp decline. For sites 2 and 3, where the wind speed profile does not exhibit drastic variations in the active power production range, the performance of the two methods are comparable.

4. Conclusions

The use of fractional-ARIMA models was proposed to model and forecast hourly average wind speeds. The choice of the model was motivated by its ability to incorporate long range correlations which were recently shown to exist in wind speed records. The unique merit of *f*-ARIMA models is their ability to parsimoniously capture time series measurements in the presence of correlations, both short term and long term. The models were applied to forecast hourly average wind speeds up to two days (48 h) ahead. The results suggest that the proposed method is able to improve the accuracy of forecasting (as measured by the $\sqrt{\text{FMSE}}$) by an average of 42% compared to the persistence method. The forecasted wind speeds were used in conjunction with the power curve of an operational wind turbine generator to obtain corresponding forecasts of the output powers. While forecast errors for wind speeds below the cut-in speed are inconsequential as far as power forecasts are concerned, the proposed method is noted to be beneficial during volatile wind speed regimes. The economic impact of errors in wind turbine output power prediction on the market integration of wind farms and a systematic characterization of different wind power prediction models would be interesting topics for further research.

Acknowledgment

The financial support from ND EPSCOR through NSF grant EPS 0132289 is gratefully acknowledged. The authors wish to thank Mr. David Kahly with Moorhead Public Service for providing the power curve data of their NEG MICON wind turbine. The authors also thank the reviewer for constructive suggestions which have helped enhance the quality of this manuscript.

Appendix A

The wind speeds (miles/hour) are recorded by means of conventional cup type anemometers located at a height of 20 m. Wind speeds acquired every 2 s are averaged over a 10 min interval to compute the 10 min average wind speed. The 10 min average wind speeds are further averaged over a period of 1 h to obtain the hourly average wind speed.

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